

Practical 4

Aim:

Working with Data structure – Part I:

- a) Write a program to remove duplicates from list.
- b) Write a program to find frequency of elements of list.
- c) Write a program to sort given list.
- d) Write a program for matrix addition and matrix multiplication using list.
- e) Write a program to generate Pascal's triangle using list.
- f) Write a program to find minimum and maximum value in list of tuples.
- g) Write a program to remove an element from tuple.

Code:

A.

```
lst = [1, 2, 2, 3, 4, 4, 5]
```

```
new = []
```

```
for i in lst:  
    if i not in new:  
        new.append(i)
```

```
print("List:", new)
```

B.

```
lst = [1, 2, 2, 3, 3, 3, 4]
```

```
for i in set(lst):  
    print(i, "=", lst.count(i))
```

C.

```
lst = [5, 2, 8, 1, 3]
```

```
lst.sort()
```

```
print("Sorted list:", lst)
```

D.

Matrix Additon

A = [[1, 2], [3, 4]]

B = [[5, 6], [7, 8]]

C = []

```
for i in range(2):  
    row = []  
    for j in range(2):  
        row.append(A[i][j] + B[i][j])  
    C.append(row)
```

```
print("Addition:")
```

```
for row in C:  
    print(row)
```

Matrix Multiplication

A = [[1, 2], [3, 4]]

B = [[5, 6], [7, 8]]

C = [[0, 0], [0, 0]]

```
for i in range(2):  
    for j in range(2):  
        for k in range(2):  
            C[i][j] += A[i][k] * B[k][j]
```

```
print("Multiplication:")
```

```
for row in C:  
    print(row)
```

E.

n = 5

triangle = []

for i in range(n):

 row = [1] * (i + 1)

 for j in range(1, i):

 row[j] = triangle[i-1][j-1] + triangle[i-1][j]

 triangle.append(row)

for row in triangle:

 print(row)

F.

lst = [(10, 20), (5, 30), (15, 2)]

values = []

for t in lst:

 values.extend(t)

print("Minimum:", min(values))

print("Maximum:", max(values))

G.

t = (10, 20, 30, 40, 50)

x = int(input("Enter element to remove: "))

lst = list(t)

lst.remove(x)

t = tuple(lst)

print("Tuple:", t)

Output:

A.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code - Python ⚠️
● PS D:\clg sem 5\Python> python -u "d:\clg sem 5\Python\para3.1.py"
List: [1, 2, 3, 4, 5]
❖ PS D:\clg sem 5\Python> 
```

B.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code - Python ⚠️
● PS D:\clg sem 5\Python> python -u "d:\clg sem 5\Python\para3.2.py"
1 = 1
2 = 2
3 = 3
4 = 1
❖ PS D:\clg sem 5\Python> 
```

C.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code - Python ⚠️
● PS D:\clg sem 5\Python> python -u "d:\clg sem 5\Python\para3.3.py"
Sorted list: [1, 2, 3, 5, 8]
❖ PS D:\clg sem 5\Python> 
```

D. Matrix Additon

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code - Py
● PS D:\clg sem 5\Python> python -u "d:\clg sem 5\Python\para.3.4.1.py"
Addition:
[6, 8]
[10, 12]
❖ PS D:\clg sem 5\Python> 
```

Output:

D. Matrix Multiplication

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code
● PS D:\clg sem 5\Python> python -u "d:\clg sem 5\Python\para3.4.2.py"
Multiplication:
[19, 22]
[43, 50]
PS D:\clg sem 5\Python> 
```

E.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code
● PS D:\clg sem 5\Python> python -u "d:\clg sem 5\Python\para3.5.py"
[1]
[1, 1]
[1, 2, 1]
[1, 3, 3, 1]
[1, 4, 6, 4, 1]
PS D:\clg sem 5\Python> 
```

F.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code - Python
● PS D:\clg sem 5\Python> python -u "d:\clg sem 5\Python\para3.6.py"
Minimum: 2
Maximum: 30
PS D:\clg sem 5\Python> 
```

G.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Code - Python
● PS D:\clg sem 5\Python> python -u "d:\clg sem 5\Python\para3.7.py"
Enter element to remove: 10
Tuple: (20, 30, 40, 50)
PS D:\clg sem 5\Python> 
```